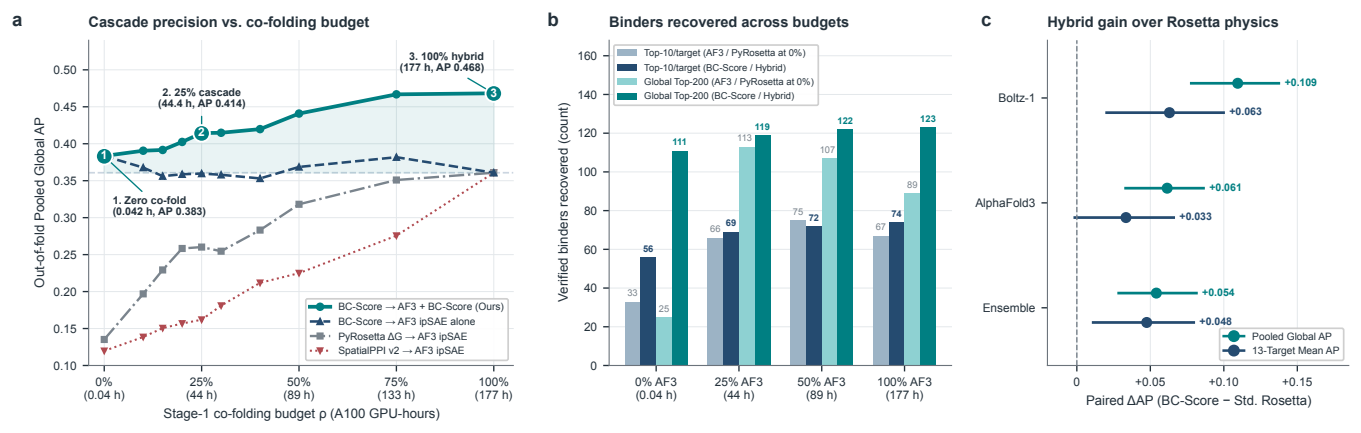

a
Cascade precision vs. co-folding budget
3. 100% hybrid
(177 h, AP 0.468)
2. 25% cascade
(44.4 h, AP 0.414)
1. Zero co-fold
(0.042 h, AP 0.383)
Out-of-fold Pooled Global AP
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0%
(0.04 h)
25%
(44 h)
50%
(89 h)
75%
(133 h)
100%
(177 h)
Stage-1 co-folding budget ρ (A100 GPU-hours)
BC-Score → AF3 + BC-Score (Ours)
BC-Score → AF3 ipSAE alone
PyRosetta ΔG → AF3 ipSAE
SpatialPPI v2 → AF3 ipSAE
b
Binders recovered across budgets
Verified binders recovered (count)
Top-10/target (AF3 / PyRosetta at 0%)
Top-10/target (BC-Score / Hybrid)
Global Top-200 (AF3 / PyRosetta at 0%)
Global Top-200 (BC-Score / Hybrid)
160
140
120
100
80
60
40
20
0
33
56
25
111
66
69
113
119
75
72
107
122
67
74
89
123
0% AF3
(0.04 h)
25% AF3
(44 h)
50% AF3
(89 h)
100% AF3
(177 h)
c
Hybrid gain over Rosetta physics
Boltz-1
AlphaFold3
Ensemble
+0.109
+0.063
+0.061
+0.033
+0.054
+0.048
Pooled Global AP
13-Target Mean AP
0
+0.05
+0.10
+0.15
Paired ΔAP (BC-Score − Std. Rosetta)